AUTOSAR High-Level Design Specification: Engine Control Unit (ECU)
Document Version: v2.0
Classification: Tata Tech Confidential

1. Executive Overview
This document specifies the updated software architecture for the Engine Control Unit (ECU) supporting hybrid p

2. Software Components
The architecture consists of the following primary software components:
- Component: EngineManagementSwc
- Component: FuelInjectionSwc
- Component: IgnitionControlSwc
- Component: BatteryManagerSwc (NEW in v2.0)
- Component: HybridTorqueSwc (NEW in v2.0)

3. Interfaces and Ports
- Interface: If_EngineSpeed (Provides engine RPM calculation signals).
- Interface: If_ThrottleCmd (Port: PPort_ThrottlePosition to RPort_ThrottleActuator).
- Interface: If_FuelInjectionTiming (Port: PPort_FuelPulseWidth).
- Interface: If_HybridTorqueReq (NEW Interface for hybrid power distribution).

4. Signal Specifications
- Signal: Sig_EngineRPM (Data type: uint16, Range: 0 - 8000 RPM)
- Signal: Sig_ThrottleAngle (Data type: float32, Range: 0.0 - 100.0 %)
- Signal: Sig_BatterySoC (NEW Signal: State of Charge, Range: 0 - 100 %)